Computers and Electrical Engineering 123 (2025) 110004

Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

SC-MambaFew: Few-shot learning based on Mamba and selective spatial-channel attention for bearing fault diagnosis

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ARTICLE INFO

Dataset link: https://github.com/giabao804/few-shot-mamba.git

Keywords:
Bearing fault diagnosis
Few-shot learning
Mamba
ConvMixer
Covariance metric neural network

ABSTRACT

Bearings are crucial yet vulnerable components in electrical equipment, prone to rapid degradation and failure. Diagnosing bearing failures is both critical and highly informative, but constructing large datasets, especially in industrial settings, remains a significant challenge. To address this issue, we propose a novel end-to-end few-shot learning framework, SC-MambaFew(Selective Channel Mamba-based Few-shot learning), for bearing fault diagnosis, which achieves high performance even with limited training data. The proposed framework is built on the Mamba architecture incorporating spatial attention and a Global-Local Channel Attention (GLCA) module to extract both fine-grained local details and broader global patterns, enhancing model performance. The extracted features are further refined through aSelective Channel (SC) mechanism and passed into a covariance-based classification module. Extensive experiments on the HUST and Case Western Reserve University (CWRU) bearing datasets, evaluated using metrics such as Accuracy, Recall, and F1 score, as well as statistical significance testing with the Wilcoxon signed-rank test, demonstrates that the proposed SC-MambaFew outperforms existing deep learning methods. Both qualitative and quantitative results highlight its robustness and potential for real-world industrial applications, particularly in data-scarce scenarios.

Our code is available at: https://github.com/giabao804/few-shot-mamba.

1. Introduction

Roller bearings play a vital role in rotating machinery, facilitating the movement and rotation of engine components. However, their performance can degrade due to wear and unevenness, leading to significant system issues such as unexpected machine shutdowns, increased energy consumption, and reduced engine efficiency. Failures of roller bearings can impact the entire production line in addition to the engine. Unplanned machine shutdowns and production pauses can cause substantial time and financial losses. Therefore, monitoring and maintaining the stable operation of critical components in factories and manufacturing plants is essential. Online machine condition monitoring systems have become increasingly important in identifying machinery issues early, thereby preventing production system inefficiencies, the risk of breakdowns, or unwarranted malfunctions.

In bearing fault diagnosis tasks, traditional methods such as the Hilbert transform [1,2] have played a significant role in extracting features from vibration signals and detecting anomalies. These techniques provide insights into oscillation amplitude, characteristic frequencies, and phase shifts, aiding in the analysis of system operating conditions. However, manual signal processing can be

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https://doi.org/10.1016/j.compeleceng.2024.110004
Received 21 September 2024; Received in revised form 19 November 2024; Accepted 8 December 2024

Available online 26 December 2024

complex, time-consuming, and resource-intensive, often requiring expert intervention. Recent advancements in artificial intelligence have introduced machine learning (ML) and deep learning (DL) models as promising tools for bearing fault diagnosis. Classical ML algorithms, including Support Vector Machine (SVM) [3], Relevance Vector Machine (RVM) [4], K-Nearest Neighbors (KNN) [5], Sparse Measure [6], Long Short-Term Memory (LSTM) [7], and Singular Value Decomposition (SVD) [8], have demonstrated varying degrees of success. Innovative combinations, such as the integration of Dempster–Shafer (DS) evidence theory with multiclass relevance vector machine (mRVM) [4], have proven effective in handling uncertain information in bearing data.

While machine learning methods show promise, the current trend favors deep learning models for their ability to automatically extract features from large and nonlinear datasets. Variations of Convolutional Neural Networks (CNNs) [9], such as temporal and dimensional attention mechanisms [10] and the Attention Mechanism Wide-scale Multichannel Temporal Convolutional Network (AWM-TCN) [11], have improved fault detection accuracy. Transfer learning techniques further enhance these models by leveraging pre-trained architectures from large datasets [12,13].

Despite their success, the performance of deep learning models heavily depends on the availability of high-quality labeled data, which is often scarce in industrial settings. Within industrial rotating machinery, accessing and collecting data from bearings located deep within motor systems poses formidable challenges due to harsh operating conditions, including high temperatures, vibrations, and pressure. To address this, few-shot learning (FSL) models have been explored. Baselines such as the Siamese Neural Network [14] have been extended with architectures like Wide-Kernel Deep CNNs (WD-CNN) by Zhang et al. [15] and ConvMixer architecture by Vu et al. [16] to address bearing fault classification with limited training data. Furthermore, the utilization of Matching Net by Li et al. [17] and the application of Prototype Network by Shen et al. [18] have been put into practice. For problems with insufficient labels, Wang et al. [19] combined self-supervised learning (SSL) with Siamese Networks to capture inherent features of unlabeled samples. Research on few-shot signal identification has demonstrated that, while outperforming conventional models, diagnosis still needs to be performed after several steps. However, these methods often rely on simplistic metrics like Euclidean and cosine distances, limiting their ability to handle complex scenarios. With a different approach, Li et al. developed an end-to-end architecture named CovaMNET [20], based on distribution consistency. With the covariance matrix acting as a representation of the correlation between each sample in the query set and the support set, several studies [21,22] across a wide range of fields have demonstrated the efficacy of this measure.

On the other hand, although Convolutional Neural Network variants have shown great success in fault diagnosis, including their ability to effectively capture local patterns in the data, these models often struggle with long-range dependencies or global context, which are essential for diagnosing more complex faults or those that manifest over longer time spans. To address the limitations of CNNs in modeling long-range dependencies, Transformer-based models have gained attention. Originally developed for natural language processing (NLP), Transformers utilize self-attention mechanisms to capture both local and global dependencies in sequences. In the realm of bearing fault analysis, to overcome weaknesses of CNN-based models, some works have been proposed based on the Transformer architecture, such as Vu et al. [23], Wang et al. [24,25], and Han et al. [26]. These studies demonstrate effectiveness due to the high impact of the "self-attention" mechanism.

Despite their performance, Transformer-based models require substantial computational resources when processing large-scale datasets or high-resolution images, resulting in a significant processing burden. A question arises about exploring a technique that would not only be an efficient feature extractor but also manage computational requirements effectively. State–space models (SSM), an important concept from control theory, have recently gained traction in deep learning. Researchers' efforts in creating S4 [27] have mitigated issues such as vanishing and exploding gradients and reduced efficiency with longer inputs. However, focusing on accuracy, S4 still has a long way to go to catch up with Transformers. Thanks to Gu and Dao's introduction of State Space Models with Selective Scan (S6), also known as Mamba [28], the technique enhanced S4 with a selective mechanism that emphasizes pertinent data in an input-dependent manner. This method forms the crucial foundation for applying SSM to image tasks. Although Mamba has been widely applied in classification tasks, such as Vision Mamba [29] and VMamba [30], as well as segmentation tasks [31–33], to the best of our knowledge, it has not been explored for few-shot bearing fault diagnosis.

Building upon the limitations of existing works, in this study, we introduce SC-MambaFew, a novel framework designed to bridge the gaps in feature extraction, model adaptability, and performance under data-scarce scenarios. The proposed approach leverages the Mamba architecture and Channel Selection (SC) that improves feature extraction by selectively focusing on the most important channels in the input data.

The motivations and contributions of this work can be outlined as follows:

- Overcoming multistage limitations with end-to-end framework: Unlike previous multistage approaches, SC-MambaFew employs an end-to-end framework, eliminating intermediate steps that can degrade performance. This unified structure simplifies deployment and ensures feature extraction and classification are jointly optimized, resulting in more robust and accurate fault diagnosis.
- Improving feature extraction: Existing methods often rely on either spatial or channel attention but fail to holistically capture both dimensions. SC-MambaFew introduces a novel dual-attention design:
  - The GL-Mamba module enhances spatial feature extraction, addressing the lack of spatial adaptiveness in many state-of-the-art methods, and innovatively applying Mamba to few-shot bearing fault diagnosis.
  - The GLCA mechanism that integrates both global and local attention to identify important features across the entire input and within specific local regions. Combined with the MLKFE block, GLCA compensates for the channel attention gap by extracting multiscale features, crucial for handling diverse fault types and operating conditions.

- Enhancing generalizability with Covariance Metric: SC-MambaFew leverages the Covariance Metric to model the distribution consistency of feature embeddings. This approach ensures robust comparisons between query images and the support set, enhancing adaptability to varying datasets and conditions.
- Combining spatial and channel insights for comprehensive representation: By integrating the Mamba Attention (a mechanism where Spatial Attention is designed based on Mamba) and GLCA mechanism, SC-MambaFew achieves richer and more comprehensive feature representations, improving diagnostic accuracy.

In summary, SC-MambaFew offers a lightweight, adaptable, and efficient solution for real-world applications in bearing fault diagnosis.

2. Related work

Conv-Mixer: The Conv-Mixer model [34] is a simple neural network with significantly lower computational costs compared to models like Vision Transformer (ViT) [35], MLP-Mixer [36], and ResNet [37], while still maintaining high performance on similar dataset sizes. Inspired by ViT and MLP-Mixer, ConvMixer divides images into patches and processes them through Depthwise Convolution (DWC) and Pointwise Convolution (PWC) blocks. This ConvMixer model enables the extraction of spatial features independently for each channel via DWC block and then combines them to create rich and detailed image representations via PWC block, thereby delivering powerful and efficient processing capabilities.

State Space Models (SSMs): are widely used in control theory to describe, analyze, and design multivariate systems. These models represent a system using a set of first-order differential equations (ODEs) that relate the input sequence $u(t) \in \mathbb{R}^{1\times D}$ to the output $v(t) \in \mathbb{R}^{1\times D}$ via a hidden state space $s(t) \in \mathbb{R}^{1\times N}$ $(N > D)$:

$$\begin{aligned} s'(t) &= \mathcal{A}s(t) + \mathcal{B}u(t), \\ v(t) &= \mathcal{C}s(t), \end{aligned} \tag{1}$$

where $\mathcal{A} \in \mathbb{R}^{N\times N}$, $\mathcal{B} \in \mathbb{R}^{N\times D}$, and $\mathcal{C} \in \mathbb{R}^{D\times N}$ represent the state transition matrix, input matrix, and output matrix, respectively. This formulation, while insightful, is expressed in continuous-time and thus difficult to directly utilize in deep learning models that operate in discrete-time steps. To adapt this for implementation in neural networks, it is discretized from its continuous-time formulation using the zero-order hold (ZOH) approach with a given time step $\Delta > 0$. The discrete approximations of $\mathcal{A}$ and $\mathcal{B}$ are computed as follows:

$$\begin{aligned} \overline{\mathcal{A}} &= \exp(\Delta\mathcal{A}) \\ \overline{\mathcal{B}} &= (\Delta\mathcal{A})^{-1}\left(\exp(\Delta\mathcal{A}) - \mathcal{I}\right)\mathcal{B} \end{aligned} \tag{2}$$

where $\mathcal{I}$ is the identity matrix. Using these discretized matrices $\overline{\mathcal{A}}$ and $\overline{\mathcal{B}}$ the state–space model can be updated in a recursive manner for each time step t:

$$\begin{aligned} s_t &= \overline{\mathcal{A}}s_{t-1} + \overline{\mathcal{B}}u_t \\ v_t &= \mathcal{C}s_t \end{aligned} \tag{3}$$

Setting the initial state $s_{-1} = 0$ for the first input ensures that the system starts without any hidden state. The output sequence can then be computed via a global convolution operation:

$$\begin{aligned} \overline{\mathcal{K}} &= \left(\mathcal{C}\overline{\mathcal{B}}, \mathcal{C}\overline{\mathcal{A}\mathcal{B}}, \ldots, \mathcal{C}\overline{\mathcal{A}}^{L-1}\overline{\mathcal{B}}\right) \\ \mathbf{v} &= \mathbf{u} * \overline{\mathcal{K}} \end{aligned} \tag{4}$$

where $\overline{\mathcal{K}} \in \mathbb{R}^{L}$ is a convolutional kernel, and L is the length of the input sequence. This kernel is precomputed from $\overline{\mathcal{A}}$, $\overline{\mathcal{B}}$, and $\mathcal{C}$, allowing for efficient parallel computation.

Mamba and Vmamba: Inspired by the State–Space Model (SSM), a classic approach in control theory used to describe and analyze multivariable systems, Gu and Dao introduced the State Space Models with Selective Scan (S6), also known as Mamba [28]. Despite being newly launched, Mamba has rapidly gained recognition from the AI research community. This innovative approach is highly promising, applicable to various Natural Language Processing (NLP) and Computer Vision (CV) tasks, and rivals Transformer-based models in terms of speed, memory efficiency, and accuracy. Liu et al. built on the capabilities of Mamba to propose Vmamba [30] for vision tasks, leveraging a 2D-Selective-scan (SS2D) module. This enhancement allows Vmamba to process 2D image data while retaining the advantages of the original Mamba architecture, such as linear complexity and global receptive fields.

2D Selective Scan: The 2D Selective Scan is an extension of the Mamba architecture, specifically designed to handle 2D image data, as opposed to the 1D sequences in NLP tasks. SS2D enhances the state–space modeling capabilities by employing a four-directional scanning mechanism, referred to as Cross-Scan, which captures spatial dependencies in image patches. The input image is divided into patches and scanned along four different directions, producing four separate sequences. These sequences are fed into the S6 block (the core module of Mamba), which processes them individually. The outputs from these four directions are then aggregated to form a 2D feature map that is passed to the next layer in the network:

$$\hat{z} = SS2D(z) = \sum_{i=1}^{4} S6\left(\text{scan}(z, i)\right) \tag{5}$$

where z is the input feature map, and $\hat{z}$ is the output feature map after passing through the SS2D block. The function $\mathrm{scan}(z, i)$ converts the patches of the input image into sequences based on the direction i. SS2D offers a linear computational complexity while maintaining global receptive fields, making it highly effective for image processing tasks that require efficient and scalable feature extraction.

Covariance Metric Network: In the realm of few-shot learning, various approaches have been explored to address the challenges of learning when only a small number of samples are available. Traditional metric learning approaches, such as Prototypical Networks [38] and Relation Networks [39], typically rely on simple distance metrics like Euclidean distance or learnable similarity measures. Li et al. proposed an end-to-end Covariance Metric Network (CovaMNet) [20] to enhance the performance of few-shot learning by exploiting covariance metrics to extract second-order statistical information from the data, thereby significantly improving the performance of few-shot learning models. Experiments have demonstrated that CovaMNet achieves good results on various datasets, showcasing significant potential for exploitation and development in few-shot learning tasks.

3. Methodology

3.1. The proposed model

Distance metrics like Euclidean and cosine similarity are foundational tools in machine learning, commonly used for tasks such as clustering, classification, and nearest neighbor searches. These metrics perform well in low-dimensional spaces where the data distribution is relatively simple. However, as the dimensionality of the data increases, these metrics encounter several challenges. In high-dimensional spaces, the notion of distance becomes less intuitive, and the distance between any two points tends to converge, a phenomenon often referred to as the "curse of dimensionality". This convergence makes it difficult to distinguish between different samples, especially when the data distribution is complex. Moreover, in cases where the available data is limited, the effectiveness of these distance metrics further diminishes. For instance, in applications involving spectrum images, where the features may be represented in only two dimensions in Euclidean space, traditional metrics may fail to capture the intricate relationships between data points. This limitation makes it challenging to build robust models that can generalize well to new, unseen data. Therefore, there is a pressing need for alternative approaches that can overcome these challenges and provide more reliable feature comparisons in high-dimensional spaces.

To address these limitations, we introduce a novel few-shot learning framework that integrates attention mechanisms with a covariance-based metric [20] for feature comparison. Our framework is designed to work effectively with high-dimensional data, such as spectrum images, where traditional distance metrics fall short. The key innovation lies in the use of both spatial attention based on Mamba [40] which are very trendy now and channel attention mechanisms, which allow the model to focus on the most relevant features, combined with a covariance metric that captures the relationships between these features more effectively than traditional distance measures.

The first step in our framework involves transforming the input data from the time domain to the spectral domain. We apply the transformation method described in [41], which utilizes the Short-time Fourier transform (STFT) to convert time-domain signals into their spectral form. This transformation is crucial for preserving the important frequency information that is often lost in traditional signal processing methods. Unlike preprocessing techniques such as high-pass or low-pass filtering, which require careful selection of cut-off frequencies and can result in data loss, our approach directly uses time–frequency graphs. This approach ensures that the model retains both time-domain and frequency-domain information, which is essential for accurate feature extraction and subsequent learning.

Our proposed SC-MambaFew framework, as shown in Fig. 1, incorporates two types of attention mechanisms: spatial attention and channel attention. These mechanisms are designed to enhance the model's ability to focus on the most informative parts of the input data, thereby improving the quality of the extracted features. The spatial attention mechanism in our framework is inspired by the Mamba architecture, which has gained popularity in recent research [28]. Spatial attention named Mamba Attention implying that the spatial Attention is built based on Mamba architecture helps the model to focus on specific regions of the input data that are most relevant to the task at hand. By highlighting these regions, the model can more effectively capture the spatial relationships within the data, which is particularly important in applications involving images or other types of structured data. In addition to spatial attention, our framework also includes the Channel Attention which is designed to focus on the most relevant channels or features. This mechanism is developed by combining both global and local attention, allowing the model to consider the importance of each feature across the entire input space as well as within specific local regions. The combination of these two types of attention mechanisms enables the model to capture a more comprehensive set of features, thereby improving its ability to differentiate between different classes.

The final component of our framework is the classification module namely Covariance Metric Layer in Fig. 1, which utilizes a Covariance-based Metric to compare features. This approach is inspired by the work of [20], which demonstrated the effectiveness of covariance matrices in capturing the relationships between features in high-dimensional spaces.

In our framework, the dataset is divided into two parts: a support set (**S**) with **K** labeled samples for each class and a query set (**Q**) with **N** distinct categories. The proposed architecture consists of two main parts: the feature extraction module and the classification module. After the input images are processed through the spatial and channel attention modules, the extracted features are combined using a channel-selective mechanism. Let c, h, and w represent the channel, height, and width of the feature maps, respectively.

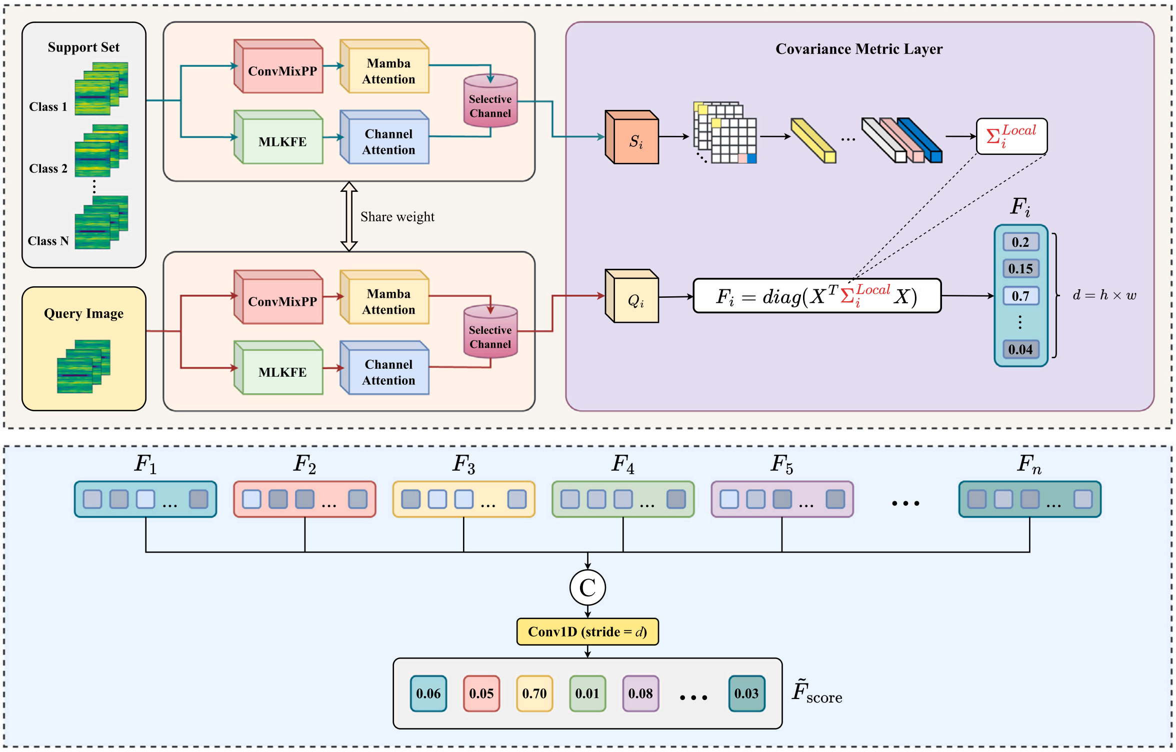


Fig. 1. Our proposed SC-MambaFew model.

Define $d = h \times w$ as the total spatial dimension of the feature map. The features from the support set are used to compute the **local covariance matrix** for each class, defined as:

$$\Sigma_i^{\text{Local}} = \frac{1}{n-1}\sum_{j=1}^{K}(X_j - \mu)(X_j - \mu)^{\top}, \tag{6}$$

where $X_j \in \mathbb{R}^{c\times d}$ represents the j-th input feature, and $\mu \in \mathbb{R}^{c\times d}$ denotes the mean vector of the features within the class. Here, $n = K \times d$ is the total number of local deep descriptors, calculated as the product of the number of samples K and the feature dimension d. The covariance matrix Σ_i^{Local} characterizes the relationships among feature dimensions for each class. By capturing these relationships, it provides a robust similarity measure that extends beyond traditional distance metrics, improving class representation and differentiation.

The relationship between a query image and the support set is then determined by the following function:

$$f(\mathbf{x}, \Sigma_i^{Local}) = diag(\mathbf{x}^T\, \Sigma_i^{Local}\mathbf{x}) \tag{7}$$

where $\mathbf{x} \in \mathbb{R}^{c\times d}$ is the feature vectors of the query image, and Σ_i^{Local} is the class-specific local covariance matrix. This metric compares the query image's features against the support set's features. The function $f(\mathbf{x}, \Sigma_i^{Local})$ computes a scalar value that indicates the correlation between the query image's feature vector and the class-specific distribution. This correlation score reflects how well the features of the query image align with the characteristics of class i. By evaluating this score across all classes in the support set, we obtain a vector that represents the degree of similarity between the query image and each class. The vector of similarity scores for all classes is then concatenated sequentially, forming a comprehensive feature vector of length $d \times N$, where d is the dimensionality of the feature vector $f(\mathbf{x}, \Sigma_i^{Local})$ and N is the total number of classes, called as F.

We utilize a one-dimensional convolutional layer to compute the statistical score for each query image in order to determine its class affiliation. The kernel size and stride for this layer are denoted by d. The computation can be formalized as follows:

$$\tilde{F}_{\text{score}} = \text{conv1d}(F, \text{stride} = d) \tag{8}$$

In this equation, $\tilde{F}_{\text{score}}$ is the statistical score calculated by the one-dimensional convolution operation applied on the feature set F, with stride equal to d. This stride value, determined by the vector's length, allows the convolutional layer to systematically cross the entire values of the vector that depicting the similarity of the query image with the support image's distribution, thereby effectively calculating the statistical score and contributing in the signal fault's classification.

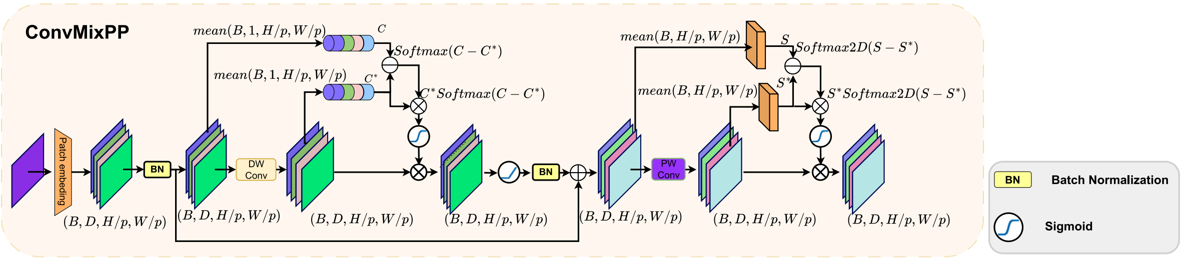


Fig. 2. ConvMixPP block.

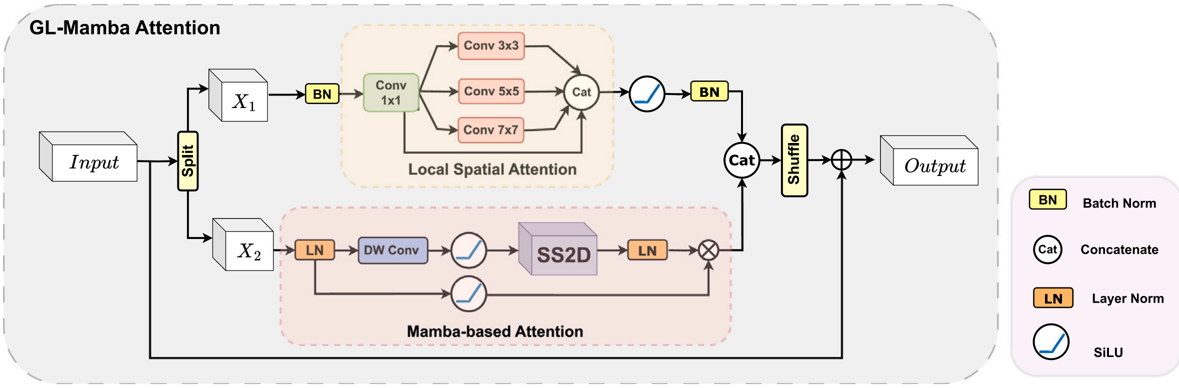


Fig. 3. GL-Mamba block. Our convolutional layers in red in local spatial attention are deployed by dilated convolutions with kernel size 3×3 and dilated rate respectively equal to 1,2,3.

3.2. Mamba attention module

While Convolutional Neural Networks (CNNs) have been widely recognized for their effectiveness in various image processing tasks, recent advancements have sought to further optimize their performance. Among these, ConvMixer [34] represents a significant breakthrough, offering a substantial reduction in computational complexity while maintaining high effectiveness in feature extraction and image classification tasks. This innovation has demonstrated that it is possible to achieve competitive performance with a more streamlined architecture. Building on the foundation laid by ConvMixer, we have utilized an enhanced version from [42], termed ConvMixer Plus Plus (ConvMixPP). Thanks to Le et al. with the Attention Mixer in [42] in proposing two core layers: Priority Spatial Attention (PSA) and Priority Channel Attention (PCA). The ConvMixPP is specifically designed to optimize the feature extraction phase, a critical step in bearing faults diagnosis task. The primary innovation in ConvMixPP, shown in Fig. 2, is inspired by Le et al. method then combining with GL-Mamba block to enhance rich information for spatial attention roles.

To obtain global attention maps efficiently, we observed that Mamba delivers highly competitive performance. Recent developments in Mamba, which are rooted in the State Space Model from control theory, have shown remarkable advancements over existing architectures, particularly the Transformer. The core mechanism of the Transformer "self-attention" requires a time complexity of $O(n^2)$ during training and $O(n)$ during inference, which can become computationally expensive as the sequence length increases. In contrast, Mamba integrates sequential processing with specific transformations that reduce algorithmic complexity, all while maintaining high performance. This efficiency, coupled with Mamba's ability to handle complex tasks effectively, motivated us to develop a module inspired by Mamba.

In our architecture, we aim to extend the spatial information captured by the ConvMixPP block to incorporate both global and local attention perspectives. This led to the creation of our novel GL-Mamba block, which is depicted in Fig. 3. Our block is developed from the SS-Conv-SSM framework proposed in [40], and it consists of two parallel branches: the local spatial branch and the main Mamba branch. The input feature maps from the ConvMixPP block, sized $\left(\frac{H}{p} \times \frac{W}{p} \times C\right)$, where p is the patch size in the Patch Embedding layer (in this case, set to 4), are divided into two equal parts along the channel dimension, with each subset being processed separately by the two branches.

In the upper branch, the focus is on capturing local spatial information at multiple scales. Similar to the method proposed in [43], the local spatial attention mechanism extracts contextual relationships by analyzing spatial patterns across different scales. Specifically, the input feature map, denoted as $\mathbf{X}_1$, with dimensions $\left(\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}\right)$, is first normalized using a BatchNorm layer. Following normalization, the data is passed through a Local Spatial Attention layer, where a 1×1 convolution reduces the number

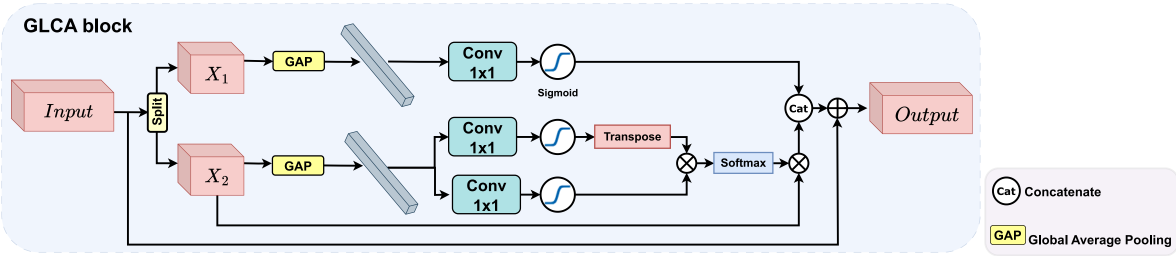


Fig. 4. GLCA block.

of channels by a factor of four, yielding a more computationally efficient tensor for subsequent processing. To capture spatial context, we apply a local attention mechanism that leverages three dilated convolutions with odd kernel sizes, ranging from 3×3 to 7×7, and dilation rates $d_i = 1, 2, 3$. These convolutions enable the model to capture both short-range and long-range dependencies within the feature map. Each of these operations can be represented as:

$$\mathbf{X}_1^{\text{reduced}} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{8}} = \text{Conv}_{1\times1}\left(\mathbf{X}_1^{\text{norm}}\right) \tag{9}$$

$$\mathbf{X}_1^{(\mathbf{d_i})} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{8}} = \text{Dilated-Conv}_{k_i \times k_i, d=d_i}\left(\mathbf{X}_1^{\text{reduced}}\right), \quad d_i \in \{1,2,3\}, \quad k_i \in \{3,5,7\} \tag{10}$$

After applying the convolutions, the resulting attention maps are activated through a SiLU non-linearity function. The output features are then concatenated along the channel dimension, yielding a tensor with dimensions $\left(\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}\right)$, which contains rich local spatial information. This process can be formalized as:

$$\mathbf{X}_1^{\text{local}} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}} = \text{Concat}\left(\mathbf{X}_1^{(\mathbf{reduced})}, \mathbf{X}_1^{(\mathbf{d_1})}, \mathbf{X}_1^{(\mathbf{d_2})}, \mathbf{X}_1^{(\mathbf{d_3})}\right) \tag{11}$$

Thus, the resulting tensor in $\mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}}$ contains spatial patterns across various scales, significantly enriching the model's local spatial information representation. The lower branch is dedicated to extracting global attention by analyzing long-range dependencies across the entire feature map. The input features in this branch undergo Layer Normalization followed by a depthwise convolution with a kernel size of 5×5, which helps to preserve spatial structure while reducing computational overhead.

$$\mathbf{X}_2^{\text{conv}} = \text{Depthwise-Conv}_{5\times5}\left(\text{LayerNorm}(\mathbf{X}_2)\right) \tag{12}$$

Next, the processed features are passed into the 2D Selective Scan layer [30], which captures global relational information across the image. Unlike the original SSM-based approach, we streamlined this branch by removing unnecessary linear layers, focusing instead on the core functionality of the SS2D layer to reduce parameter count and improve efficiency. With SS2D, the model scans global information through every pixel in each spectrum image in four directions: from top to bottom, bottom to top, left to right, and right to left. This multidirectional scanning allows for a thorough spatial analysis, capturing relationships between pixels across the entire feature map. After passing through the SS2D layer, the output features are normalized again via Layer Normalization and subsequently multiplied element-wise with the initial, normalized input features from this branch. This process ensures that the model effectively captures global relationships while retaining the original input's structural integrity:

$$\mathbf{X}_2^{\text{global}} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}} = \mathbf{X}_2^{\text{SS2D}} \times \mathbf{X}_2^{\text{conv}} \tag{13}$$

The final step in the GL-Mamba block involves concatenating the outputs from the local and global branches along the channel dimension. This concatenation fuses both local and global spatial information, providing the model with a comprehensive understanding of the input feature map. After concatenation, the channels are shuffled to promote interactions between different feature dimensions, further enhancing the model's representation power:

$$\mathbf{X}^{\text{final}} = \text{Shuffle}\left(\text{Concat}\left(\mathbf{X}_1^{\text{local}}, \mathbf{X}_2^{\text{global}}\right)\right) \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times C} \tag{14}$$

By combining both local and global spatial attention, the GL-Mamba block significantly improves the model's ability to capture spatial relationships, leading to better performance in classification tasks.

3.3. Channel attention module

From the fact that image data type which is very difficult to classify such as spectrum image, because of noise from the errors in the measurement stages, from transformations when converting from time-series domain to frequency domain, in early-stages, feature maps play an important role for the high performance of the model. We found that obtain information from many different directions of the images provides critical support (see Fig. 4).

In addition to the Mamba attention module, which excels at capturing spatial relationships, we recognized the need to also focus on the relationships along the channel dimension to provide a more comprehensive view of the data. To address this, we designed a new module called the Channel Attention Module, which is composed of two sub-modules: Multiscale Large Kernel Feature Extraction (MLKFE) [23] and Global Local Channel Attention (GLCA). The feature maps extracted by the MLKFE block are subsequently fed into the GLCA module, where they are further refined to capture more concentrated and relevant information.

The MLKFE [23] module is pivotal in our architecture because it enhances the model's ability to capture diverse and characteristic information, which is especially critical when dealing with limited training samples. We drew inspiration from the work of Vu et al. [23], who introduced the residual large kernel attention (ResLKA) layer. This layer is adept at capturing both short-range and long-range pixel relationships within feature maps, an attribute that is particularly important for extracting features in complex and noisy datasets, such as those used in bearing fault diagnosis. The intricate nature of these datasets, which often contain a mix of varied and noisy data, makes it essential to use an approach like MLKFE that can capture fine-grained details and more comprehensive relationships needed for accurate identification.

Building on the success of global-local attention mechanisms, particularly the one proposed by Song et al. in their Global-Local Attention Module (GLAM) [43], we introduce the Global Local Channel Attention (GLCA) module for the first time in the context of bearing fault diagnosis. Song et al.'s GLAM was designed to capture both local and global attention across spatial and channel dimensions. From this early research, recognizing the unique needs of bearing fault identification we adapted their approach by designing a specialized GLCA module focused solely on channel attention. In our GLCA module, we emphasize the significance of both local and global channel information by splitting the feature maps obtained from the MLKFE [23] backbone into two equal parts. The first part is processed by the upper branch, which is dedicated to local channel attention. This branch focuses on capturing fine-grained channel relationships that are crucial for identifying subtle patterns within the data. The second part flows through the lower branch, which is designed to capture global channel attention. This branch is essential for understanding the broader, more holistic relationships across the entire feature map. By combining the strengths of these two parallel branches, our GLCA module ensures that the model can effectively balance the need for both detailed local information and broad global context. This dual attention mechanism is particularly beneficial in bearing fault diagnosis, where the datasets are often characterized by their complexity and the presence of noise. The GLCA module enhances the model's ability to accurately identify and classify faults, even when only a limited amount of training data is available.

Similar to the Mamba attention module, to lever up the importance of both local and global information according to the deep dimension, we propose a novel channel attention mechanism that balances both local and global information through a dual-branch architecture. The input feature maps $\mathbf{X} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times C}$ from the Multiscale Large Kernel Feature Extraction (MLKFE) block are split into two parts: $\mathbf{X_1} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}}$ and $\mathbf{X_2} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}}$, where $\mathbf{X_1}$ focuses on local channel relationships and $\mathbf{X_2}$ emphasizes global interactions. In the first branch, we calculate the local channel relationships using $\mathbf{X_1}$. The process begins with Global Average Pooling (GAP) applied across the spatial dimensions to aggregate information into a channel-wise vector:

$$\mathbf{F_1} = \mathrm{GAP}(\mathbf{X_1}) = \frac{1}{\frac{H}{p} \times \frac{W}{p}} \sum_{i=1}^{\frac{H}{p}} \sum_{j=1}^{\frac{W}{p}} \mathbf{X_1}(i,j) \tag{15}$$

This produces $\mathbf{F_1} \in \mathbb{R}^{\frac{C}{2}}$, a compact representation of the channel information in $\mathbf{X_1}$. Next, to model local dependencies between channels, we apply a 1D convolution with a kernel size $k = 3$, followed by a sigmoid activation function:

$$\mathbf{F}_1' = \sigma(\mathrm{Conv1D}(\mathbf{F_1}, k = 3)) \tag{16}$$

Here, $\mathbf{F}_1' \in \mathbb{R}^{\frac{C}{2}}$ denotes the local attention scores for each channel, where σ is the sigmoid function. This process refines the local feature representation by emphasizing meaningful cross-channel interactions.

The final locally-attended feature map $\mathbf{A_1}$ is computed by an element-wise multiplication between the input $\mathbf{X_1}$ and the attention vector $\mathbf{F}_1'$, followed by a residual connection to retain the original input:

$$\mathbf{A_1} = \mathbf{X_1} \odot \mathbf{F}_1' + \mathbf{X_1} \tag{17}$$

This residual connection preserves key information from the original features, ensuring stable gradient flow during backpropagation and avoiding feature degradation.

The second branch is designed to capture global channel relationships using a non-local attention mechanism. Starting with the feature map $\mathbf{X_2} \in \mathbb{R}^{\frac{H}{p} \times \frac{W}{p} \times \frac{C}{2}}$, we first apply Global Average Pooling (GAP) to obtain a channel-wise vector:

$$\mathbf{F_2} = \mathrm{GAP}(\mathbf{X_2}) \tag{18}$$

Then, we compute two key elements, the **query** $\mathbf{Q_2} \in \mathbb{R}^{\frac{C}{2}}$ and **key** $\mathbf{K_2} \in \mathbb{R}^{\frac{C}{2}}$, by processing $\mathbf{F_2}$ through a 1D convolution with kernel size $k = 3$, followed by sigmoid activation:

$$\mathbf{Q_2} = \sigma(\mathrm{Conv1D}(\mathbf{F_2}, k = 3)), \quad \mathbf{K_2} = \sigma(\mathrm{Conv1D}(\mathbf{F_2}, k = 3)) \tag{19}$$

Simultaneously, the feature map $\mathbf{X_2}$ is reshaped into a matrix $\mathbf{V_2} \in \mathbb{R}^{\frac{HW}{p^2} \times \frac{C}{2}}$, where each row corresponds to a spatial position, and the columns represent channel activations. To model global interactions, we compute the outer product of $\mathbf{Q_2}$ and $\mathbf{K_2}$, generating a global attention map $\mathbf{F_2} \in \mathbb{R}^{\frac{C}{2} \times \frac{C}{2}}$:

$$\mathbf{F_2} = \mathrm{Softmax}({\mathbf{K_2}}^{\top}\mathbf{Q_2}) \tag{20}$$

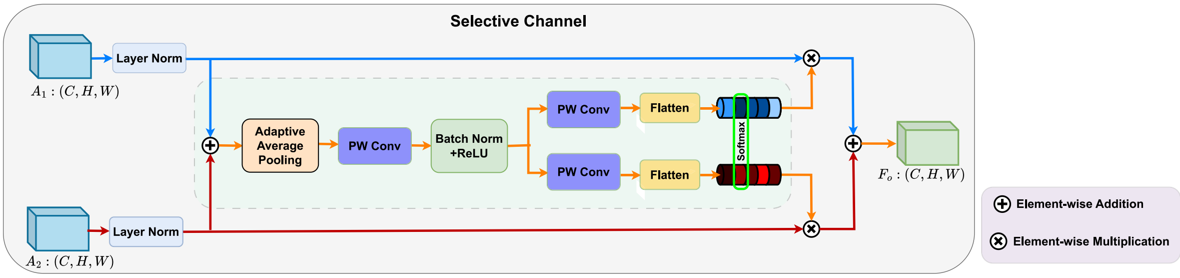


Fig. 5. Selective channel module.

The softmax function normalizes the attention map across channels, highlighting the most important global channel dependencies. The value tensor $\mathbf{V_2}$ is then modulated by the attention map $\mathbf{F_2}$ through matrix multiplication:

$$\mathbf{A_2} = \mathbf{V_2} \times \mathbf{F_2} \tag{21}$$

Finally, $\mathbf{A_2}$ is reshaped back into its original dimensions $\mathbb{R}^{\frac{H}{p}\times\frac{W}{p}\times\frac{C}{2}}$, producing the globally-attended feature map.

The final output of the attention module is obtained by concatenating the locally- and globally-attended feature maps $\mathbf{A_1}$ and $\mathbf{A_2}$ along the channel dimension:

$$\mathbf{A}^{\text{final}} = \text{Concat}(\mathbf{A_1}, \mathbf{A_2}) \in \mathbb{R}^{\frac{H}{p}\times\frac{W}{p}\times C} \tag{22}$$

This dual-branch attention mechanism efficiently captures both local and global channel dependencies, leading to improved feature representation for complex tasks such as bearing fault diagnosis.

This approach not only captures global and local channel interactions more efficiently but also maintains the stability of the gradient during backpropagation, ensures that the model remains computationally feasible, coming to more robust model efficiency in challenges requiring intricate feature representation, such as bearing fault diagnosis.

3.4. Selective channel

Inspired by the groundbreaking architecture of "Selective Kernel Networks" [44], we propose a novel module named "Selective Channel" (SC), as illustrated in Fig. 5 , as a mechanism that improves feature extraction by selectively focusing on the most important channels in the input data, ensuring that the most relevant aspects are prioritized for the task. The SC module is designed to enhance the selection process of critical channels after the input spectrum image is processed through two parallel branches, each employing a distinct attention mechanism. These branches generate two sets of feature maps, denoted as $\mathcal{A}_1$ and $\mathcal{A}_2$, which are crucial for capturing different aspects of the input data, according to Mamba attention and Channel attention.

The primary objective of the SC module is to efficiently determine the importance of each channel across the two subsets of feature maps. This begins with an element-wise summation of the outputs from the two branches, which serves to integrate the diverse information captured by each attention mechanism:

$$\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2 \tag{23}$$

This fusion step allows the model to combine the complementary strengths of the two branches, ensuring that both local and global features are preserved in the resulting feature map $\mathcal{A}$. The summation also simplifies subsequent processing by reducing the dimensionality of the problem to a single unified feature map.

To further capture the global context and channel-wise importance, we apply Global Average Pooling (GAP) to the combined feature map $\mathcal{A}$. This operation effectively compresses the spatial dimensions, producing a channel-wise statistical vector $\mathbf{g} \in \mathbb{R}_c$:

$$\mathbf{g} = \mathcal{F}_{gap}(\mathcal{A}) = \frac{1}{\frac{H}{p}\times\frac{W}{p}} \sum_{i=1}^{\frac{H}{p}} \sum_{j=1}^{\frac{W}{p}} \mathcal{A}_{(i,j)} \tag{24}$$

The GAP operation is essential because it aggregates information across the entire spatial domain, resulting in a summary statistic for each channel. This summary is crucial for understanding the overall significance of each channel, independent of the specific spatial locations.

To enable precise and adaptive channel selection, the channel-wise statistics vector $\mathbf{g}$ is then passed through a fully connected (FC) layer. This layer not only transforms the vector but also reduces its dimensionality, which is beneficial for computational efficiency. The result is a compact feature vector $\mathbf{s} \in \mathbb{R}^{C\times1}$:

$$\mathbf{s} = \zeta(\sigma(\mathbf{G})) \tag{25}$$

(a) α_c vector

(b) β_c vector

Fig. 6. Attention-weighted vectors.

where ζ denotes the activation function, which in our case is the ReLU nonlinear function, chosen for its ability to introduce non-linearity and prevent vanishing gradients. The σ term represents Batch Normalization. The compact vector $\mathbf{G}$ is derived from $\mathbf{g}$ through a 1D convolution with a kernel size $k = 1$:

$$\mathbf{G} = \mathrm{Conv1d}(\mathbf{g}, k = 1) \tag{26}$$

This step ensures that the vector $\mathbf{s}$ encapsulates the most critical channel-wise information in a compact and efficient manner.

Once we have $\mathbf{s}$, it contains two sets of values that correspond to the importance of each channel in the respective feature maps $\mathcal{A}_1$ and $\mathcal{A}_2$. To achieve an adaptive selection across different spatial scales of information, we employ a soft attention mechanism, which is guided by the compact feature descriptor $\mathbf{s}$. The soft attention mechanism applies a softmax function to the channel-wise values:

$$\alpha_c = \frac{\exp(s_u)}{\exp(s_u) + \exp(s_l)} \quad ; \quad \beta_c = \frac{\exp(s_l)}{\exp(s_u) + \exp(s_l)} \tag{27}$$

Here, α_c and β_c represent the attention weights (the effectiveness of this approach is visually illustrated in Fig. 6 with the weight of α_c vector and β_c vector) assigned to each channel c in $\mathcal{A}_1$ and $\mathcal{A}_2$, respectively. The terms s_u and s_l are the channel-wise values derived from the compact feature descriptor $\mathbf{s}$. The softmax function is crucial as it ensures that the attention weights are normalized and sum to one, which allows the network to focus on the most relevant channels while suppressing the less important ones.

Finally, the output feature map is computed by applying the attention vectors to the original feature maps. This is done through an element-wise multiplication of the attention weights α_c and β_c with the corresponding feature maps $\mathcal{A}_1$ and $\mathcal{A}_1$, followed by a summation:

$$\mathcal{F}_{out} = \alpha_c \times \mathcal{A}_1 + \beta_c \times \mathcal{A}_2 \tag{28}$$

The resulting feature map $\mathcal{F}_{out}$ effectively integrates the most relevant spatial and channel-wise information, preserving the strengths of both attention mechanisms while mitigating their individual weaknesses.

3.5. Summary on key modules of SC-MambaFew

The following section highlights the distinctive contributions of the GL-Mamba, GLCA, and SC modules, which are central to the SC-MambaFew framework.

The GL-Mamba module introduces a dual-branch mechanism that combines local spatial attention and global spatial attention to capture diverse spatial dependencies. The local spatial branch utilizes multi-scale dilated convolutions to extract fine-grained spatial patterns at varying receptive fields, ensuring sensitivity to both short- and long-range dependencies. The global spatial branch employs depthwise convolutions to efficiently model global spatial relationships. By integrating these two perspectives, GL-Mamba provides a more comprehensive spatial feature representation compared to existing methods, which often prioritize only one type of spatial attention.

The GLCA module is the first adaptation of a global-local channel attention mechanism for bearing fault diagnosis. It employs a two-branch approach to separately process local and global channel information, addressing the need for detailed feature extraction and broad contextual understanding. The local channel branch captures fine-grained channel-specific patterns critical for detecting subtle fault characteristics, while the global channel branch models holistic channel relationships across the feature map. This dual-channel focus enhances the model's ability to handle noisy and complex industrial datasets.

The SC (Selective Channel) module introduces a novel mechanism for selectively weighting feature channels based on their relevance to the diagnosis task. It integrates outputs from the GLCA branches using element-wise summation, followed by an attention-based selection process that dynamically refines feature importance. This mechanism allows the model to prioritize the most informative channels, improving diagnostic accuracy, particularly in data-scarce scenarios.

These innovations collectively address critical gaps in existing few-shot fault diagnosis methods, enabling the proposed SC-MambaFew framework to achieve superior performance under limited data conditions.

Algorithm 1 Training algorithm for our proposed framework SC-MambaFew

Require: Support Set $\mathcal{S} = \{(X_i^s, Y_i^s)\}_{i=1}^{n_s}$, Query Set $\mathcal{Q} = \{(X_j^q, Y_j^q)\}_{j=1}^{n_q}$,
Base Model f_θ initialized with random weights,
Number of epochs E, Number of iterations T, Number of classes N
Ensure: An optimized weight parameter θ
Class Probabilities $p(Y^q|X^q, \mathcal{S})$ for each example X^q in $\mathcal{Q}$

```
Initialize Random initial weights of the model θ
for e = 1 to E do
  for t = 1 to T do
    Form a support set S = (X_i, Y_i)_{i=1}^{n_s} and a query set Q = (X_i, Y_i)_{i=1}^{n_q} from the training set Δ_train, where n_s = N ∗ K and n_q = N ∗ Q
    for i = 1 to N do
      /* Feature extraction */
      Using both branches: GL-Mamba Attention and Channel Attention modules to compute feature maps of K support images at layer ith:
      F_si ← (1/K) Σ_{j=1}^{K} F_sij with F_si = [F_si1, F_si2, ..., F_siK] (from the upper branch),
      F_s'i ← (1/K) Σ_{j=1}^{K} F_s'ij with F_s'i = [F_s'i1, F_s'i2, ..., F_s'iK] (from the lower branch),
      and feature maps of query images at layer ith: F_qi and F_q'i
      /* Selective channel processing */
      Compute F'_si and F'_qi using the Selective Channel module:
      F'_si = Fusion(F_si, F_s'i) ; F'_qi = Fusion(F_qi, F_q'i)
      /* Covariance Metric Layer */
      Calculate Σ^i_local using Eq (6)
    end for
    Calculate relation vector f(F'_qi, Σ_i^Local) using Eq (7)
    Compute Ŷ^q from the statistic score F̂_score using Eq (8)
    Compute the Contrastive Loss L_{CL,S_c}(Ŷ^q, Y^q)
    /* Gradient descent update */
    Update θ'_i = θ − α∇_{θ_i} L_{CL,S_c}(f_θ)
    /* Weight update */
    θ ← θ − α∇_θ Σ_{(X,Y)∈Q} L_CL(P(X), Y)
  end for
end for
```

4. Experiment

4.1. Datasets

To evaluate the performance of the proposed SC-MambaFew method in comparison with other state-of-the-art models in bearing fault diagnosis, we conducted experiments on two public datasets, including the HUST bearing dataset [45], and the Case Western Reserve University (CWRU) dataset [46]. These datasets were selected for their relevance and distinct characteristics in fault diagnosis, allowing for a comprehensive evaluation of our model's performance, as described more detail in the following sections.

4.1.1. HUST bearing dataset

The HUST bearing dataset [45] was developed by the research team at Hanoi University of Science and Technology, downloaded at: (HUST benchmark).[1] The HUST dataset places greater emphasis on the diversity of faults and their interrelationships rather than focusing on the size of the faults. The majority of the defects in this dataset are artificially generated, each with a uniform width of 0.2 mm. The data is structured with 7 distinct labels, representing various types of faults. Three of these labels correspond to major faults that are comparable to those in the CWRU dataset, while the remaining three represent related, yet distinct, fault types. By offering a broader spectrum of fault types and their associations, the HUST bearing dataset provides an ideal platform for more comprehensive fault diagnosis and classification tasks. This diversity makes it highly suitable for evaluating the effectiveness of machine learning models in capturing complex fault patterns and relationships.

[1] https://data.mendeley.com/datasets/cbv7jyx4p9/3.

4.1.2. CWRU dataset

The Case Western Reserve University (CWRU) dataset [46] is widely regarded as a benchmark in machine learning tasks, particularly in fault detection and bearing fault diagnosis. The CWRU data can be downloaded at: (CWRU benchmark).[2]

This dataset, with labeled data, is commonly employed for training, testing, and algorithm development in these fields. The CWRU dataset is organized into four main categories: 48k normal-baseline, 48k drive-end fault, 12k drive-end fault, and 12k fan-end fault. Each fault class is further categorized by three fault locations: Ball fault, Inner ring fault, and Outer ring fault. Additionally, for each fault type, the dataset provides three fault diameters: 0.007 in., 0.014 in., and 0.021 in. This detailed structure of the dataset allows for a thorough evaluation of models across a range of fault conditions and is frequently used as a standard for benchmarking in the bearing fault diagnosis community.

4.2. Implementation details

Before initiating the training process, we divide the dataset into query-support pairs. This strategy allows the few-shot end-to-end approach to effectively utilize the available information, even when data is limited. The support set is split into several image classes, and during each training iteration, the number of images required is determined by the number of classes in the dataset. For example, in the case of 1-shot learning, the required number of images is calculated as $1 \times \text{num\_classes} + 1$, while for 5-shot learning, it becomes $5 \times \text{num\_classes} + 1$. We use the HUST Bearing dataset with 7 classes and the CWRU dataset with 10 classes. During training, ContrastiveLoss is employed as the loss function for our proposed model, ensuring effective learning from the available data. The training process of our proposed model is given in Algorithm 1.

To ensure a fair comparison between our proposed model and other state-of-the-art approaches, we standardized conditions across all methods. For each method, we applied an Adam optimizer and used a learning rate scheduler that reduces the learning rate by a factor of 2 every 10 epochs during training with the init value is 0.001. In experiments, we trained all compared methods for 100 epochs with NVIDIA TeslaT4 16 GB GPU.

The total number of test samples in our experiments for the HUST dataset is 700 samples, and 750 samples for the CWRU dataset. The maximum number of training samples for the HUST dataset is 16 800, and for the CWRU dataset, it is 19 800. Various experimental scenarios with limited data were conducted, including 168, 840, 16 800 training samples for HUST; and 30, 60, 300, 19 800 training samples for CWRU data. In each training scenario, the number of validation samples is equal to the used testing samples with every specific dataset.

4.3. Experimental results

This section presents the experimental results for bearing fault diagnosis using the proposed SC-MambaFew model, along with recent deep learning-based few-shot models, on the HUST Bearing and CWRU datasets. To evaluate the performance of the proposed method compared to other models, we use the confusion matrix, t-SNE visualization, and evaluation metrics, including Accuracy, Recall, and F1 scores, for each test set of the HUST and CWRU datasets. It is worth noting that, due to the balanced nature of the test data setup, the Precision metric is not included in the tables as it is identical to the Accuracy. In addition, for a seasonal comparison, in tables where both 1-shot and 5-shot cases are presented, for the CWRU data, we provide results using at least 60 training samples to evaluate the model's ability in the 5-shot, 10-way setting. In the ablation study focusing solely on the 1-shot case, we include results with as few as 30 training samples to demonstrate the performance of the proposed model under conditions of very limited data.

Moreover, to demonstrate the statistical significance of the results obtained by the proposed method compared to each baseline model, we compute the p-value using the Wilcoxon signed-rank test [47]. The test is performed on the predicted probabilities for the corresponding labels, obtained from the softmax outputs of the proposed and compared models. If p-value is less than or equal to the significance level ($\alpha = 0.05$), the results are considered statistically significant indicating that the observed differences are likely genuine [48].

To be more specific, we provide statistic scores and visualization results, including plots of confusion matrix and t-distributed Stochastic Neighbor Embedding (t-SNE) of representative models on the two datasets. For quantitative assessment, we present statical metrics of the proposed model and recent deep learning- based few-shot learning models in tables, using method reference numbers, arrows for indicators, and **boldface** to highlight the best performing metrics. An upward arrow (↑) indicates that a higher value presents better performance, whereas a downward arrow (↓) signifies that a smaller value is preferred. In particular, a small p-value in a row corresponding to a method indicates that the proposed SC-MambaFew model is statistically significantly different from that method as determined by the Wilcoxon signed-rank test. Additionally, the number of parameters (Params in millions), and the Floating-Point Operations per Second (FLOPS in Gigaflops) of each model or baseline are provided in all tables. Smaller values of Params and FLOPS reflect better performance in terms of computational complexity.

The key results for both the HUST and CWRU datasets using the proposed SC-MambaFew method are summarized in Table 1. As shown in the table, even with as few as 168 training samples on the HUST dataset, the proposed model achieves an Accuracy of 86.15%, a Recall of 87.89%, and an F1 score of 87.7% in the 1-shot scenario. The results for the CWRU dataset are similarly impressive, with all evaluation metrics exceeding 97.7% even with as few as 60 training samples.

A comparison of the proposed method with other methods, along with relevant descriptions, is provided in the following sections.

[2] https://engineering.case.edu/bearingdatacenter/download-data-file.

Table 1
Key results on the HUST and CWRU datasets with different training samples by the proposed model.

(a) Key results on the HUST data with different training samples

Training samples	Test samples	Accuracy↑		Recall↑		F1↑	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
168	700	86.15	87.52	87.89	87.52	87.70	87.52
840	700	98.71	98.86	98.32	97.47	98.59	98.16
16 800	700	99.76	100	99.43	100	99.59	100

(b) Key results on the CWRU data with different training samples

Training samples	Test samples	Accuracy↑		Recall↑		F1↑	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
60	750	97.73	98.40	97.74	98.25	97.73	98.32
300	750	99.73	99.87	99.47	99.47	97.48	97.75
19 800	750	99.80	99.91	99.87	100	99.87	99.95

4.3.1. Comparison results on the HUST bearing dataset

This section presents intensive experiments conducted on the HUST bearing dataset. These experiments utilized different portions of the training data (1%, 5%, and 100%), corresponding to 168, 840, and 16 800 training samples, respectively. To evaluate the performance of the proposed SC-MambaFew model, we also re-implemented several recent deep learning-based few-shot learning methods. The compared models include CovaMNet [20] (Code),[3] SA-CovaMNet [21] which we implemented ourselves based on the paper description and the CovaMNet baseline, and MF-Net [49] (Code).[4]

Table 2 presents the results of the proposed SC-MambaFew model as well as the aforementioned models, for both the 1-shot and 5-shot scenarios, with varying amount of training samples. The results in Table 2 highlight the robustness of our approach across various training sample sizes. Notably, with only 1% of the training data (168 training samples), our method achieved 86.15% accuracy for the 1-shot case and 87.52% for the 5-shot case, outperforming the original CovaMNET [20], which only reached 80.03% for 5-shot. Similarly, our model surpassed SA-CovaMNET [21] (80.15%) and MF-Net [49] (80.10%), both of which incorporate the attention mechanism, showcasing that our Mamba-based model can outperform traditional attention mechanisms. For the case of 16 800 training samples (100% training data of the HUST dataset), the proposed SC-MambaFew also obtained comparative accuracy excepting for the 1-shot case while compared to the MF-Net (99.76% vs. 99.98%). Similar to Accuracy metric, our proposed model also show better scores in terms of Recall and F1 metrics in all training scenarios for both 1-shot and 5-shot tests.

In addition to the classification scores including Accuracy, Recall, and F1, which are scalar values computed from the entire set of predicted labels versus true labels, we calculate the p-value using the Wilcoxon signed-rank test [47] on the predicted probabilities for the corresponding labels by the proposed model and each of the aforementioned models. The p-values for the significance test between our proposed model and CovaMNET [20], SA-CovaMNET [21], and MF-Net [49] are provided in the same row as the corresponding method in Table 2. As shown in the last two columns of Table 2, except for the 5-shot case when compared with MF-Net, the p-values are smaller than 0.05, particularly in the 1-shot case compared with MF-Net ($p = 0.0009$), CovaMNet ($p \ll 0.05$ for all cases), and SA-CovaMNet ($p = 0.0003$) for both 840 and 16 800 training samples.

For further comparative interpretation, we present representative plots of confusion matrix and t-SNE for the proposed and CovaMNet and SA-CovaMnet methods, using as few as 168 training samples, in Fig. 7. The right plot of this figure (Fig. 7 c) clearly demonstrates better clustering between the regions of each category by the proposed model while compared to compared methods.

4.3.2. Comparison results on the CWRU dataset

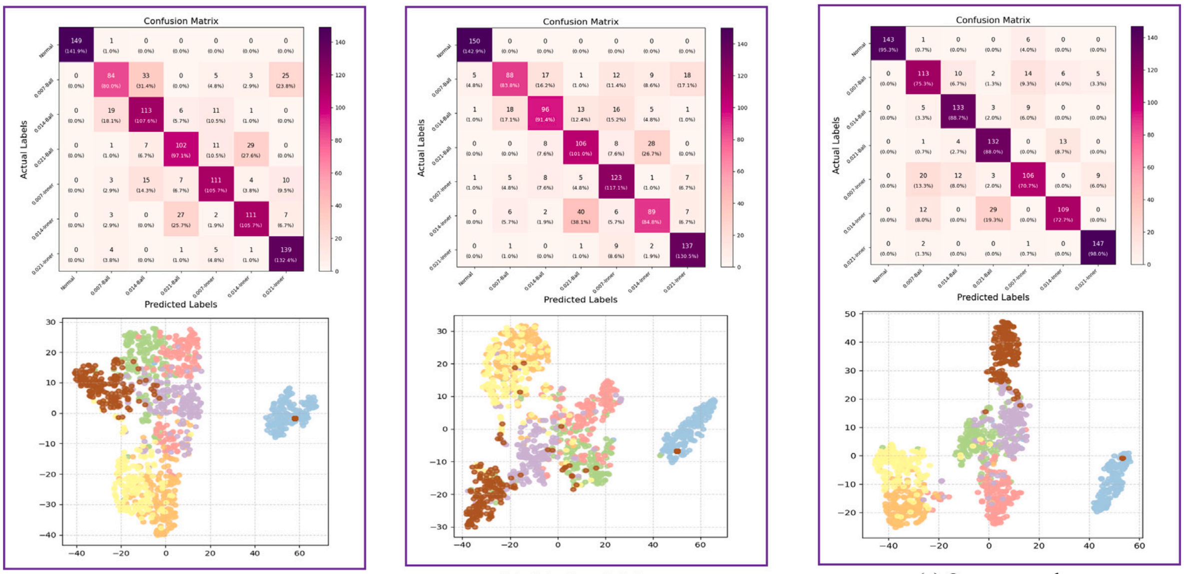
In this section, we provide a comprehensive comparison between our proposed model and several leading few-shot learning models widely used for classification tasks, including CovaMNet [20], SA-CovaMNet [21], MF-Net [49], QS-Former [24], and Ensemble-Net [23]. For a fair evaluation, we re-implemented all competing models using the open-source code provided by their respective authors except for SA-CovaMNet. The codes include: CovaMNet [20] (Code),[5] MF-Net [49] (Code),[6] QS-Former [24] (Code),[7] and Ensemble-Net [23] (Code).[8] For SA-CovaMNet [21], we implemented the model ourselves based on the paper description and the CovaMNET baseline. The experiments were conducted under both 1-shot 10-way and 5-shot 10-way scenarios, with the training data size ranging from as few as 30 samples to 19 800 samples.

Table 3 presents the evaluation metrics for both 1-shot and 5-shot experiments across varying amounts of training samples. As shown in the table, the proposed SC-MambaFew consistently outperforms the state-of-the-arts models under various experimental conditions, achieving the highest accuracy in most cases. These results underscore the robustness and generalization capability of

[3] https://github.com/WenbinLee/CovaMNet.
[4] https://github.com/HungVu307/mixer-former-fewshot.
[5] https://github.com/WenbinLee/CovaMNet.
[6] https://github.com/HungVu307/mixer-former-fewshot.
[7] https://github.com/SissiW/QSFormer.
[8] https://github.com/HungVu307/Few-shot-via-ensembling-Transformer-with-Mahalanobis-distance.

Table 2
Comparison on the HUST dataset for **1-shot** and **5-shot** cases using (a) 168 training samples, (b) 840 training samples, and (c) 16 800 training samples. The *p*-value represents the results of Wilcoxon signed-rank test between the proposed method and the compared methods.

(a) 168 training samples										
Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	76.87	80.03	78.63	80.09	79.32	80.06	0.0068	0.0020
SA-CovaMNet [21]	0.160	15.87×10^{-3}	77.88	80.15	78.63	77.81	78.25	78.96	0.0419	0.0120
MF-Net [49]	0.062	14.77×10^{-3}	78.23	80.10	78.23	77.81	78.23	78.93	0.0009	0.0980
Proposed	0.140	13.38×10^{-3}	**86.15**	**87.52**	**87.89**	**87.52**	**87.70**	**87.52**		
(b) 840 training samples										
Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	97.24	97.98	97.87	94.66	97.92	96.29	0.0006	0.0003
SA-CovaMNet [21]	0.160	15.88×10^{-3}	97.36	98.27	98.27	96.78	98.27	97.52	0.0025	0.0003
MF-Net [49]	0.062	14.77×10^{-3}	97.58	98.13	97.64	94.66	97.88	96.36	0.0043	0.0513
Proposed	0.140	13.38×10^{-3}	**98.71**	**98.86**	**98.32**	**97.47**	**98.59**	**98.16**		
(c) 16 800 training samples										
Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	98.91	99.05	98.92	97.81	98.91	98.42	0.0003	0.0293
SA-CovaMNet [21]	0.160	15.88×10^{-3}	99.01	99.85	99.01	98.89	99.01	99.37	0.0289	0.0003
MF-Net [49]	0.062	14.77×10^{-3}	**99.98**	**100**	98.56	**100**	99.26	100	0.0380	0.1168
Proposed	0.140	13.38×10^{-3}	99.76	**100**	**99.43**	**100**	**99.59**	**100**		



(a) CovaMNet (b) SA-CovaMNet (c) Our proposed

Fig. 7. Representation for 10-way accuracy classification on HUST dataset using different models with 168 training samples.

our model, particularly in scenarios with scarce data, where it still delivers outstanding scores including Accuracy, Recall, and F1 scores despite the minimal number of training samples. For instance, our proposed model demonstrates remarkable performance, achieving an accuracy of 98.40% with only 60 training samples, and further improving to 99.91% when utilizing the full dataset of 19 800 samples. This demonstrates that our model is not only effective but also highly adaptable to low-data environments, making it a strong candidate for real-world applications with limited data availability.

For pairwise model comparison using the Wilcoxon signed-rank test, the *p*-values for comparison between the proposed model and the aforementioned methods are smaller than 0.05 in most cases, especially with 19 800 training samples ($p = 0.0002$ when evaluating with QS-Former, $p = 0.0019$ when compared with Ensemble-Net and SA-CovMNet). The results indicate that the observed differences are statistically significant and unlikely due to chance, further highlighting the superior performance of the proposed SC-MambaFew compared to other state-of-the-art models.

Table 3
Comparison on the CWRU dataset for **1-shot** and **5-shot** cases using (a) 60 training samples, (b) 300 training samples, and (c) 19 800 training samples. The *p*-value represents the results of Wilcoxon signed-rank test between the proposed method and the compared methods.

Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
(a) 60 training samples										
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	92.16	93.41	92.20	93.33	92.16	93.37	0.013	0.039
SA-CovaMNet [21]	0.160	15.88×10^{-3}	92.56	92.01	92.99	95.00	92.77	93.48	0.009	0.031
MF-Net [49]	0.062	14.77×10^{-3}	94.13	95.61	93.17	95.93	93.64	95.77	0.085	0.025
QS-Former [24]	0.175	14.06×10^{-3}	92.14	94.11	83.82	93.56	87.78	93.83	0.012	0.042
Ensemble-Net [23]	0.147	14.52×10^{-3}	96.82	98.23	94.82	97.6	95.81	97.91	0.037	0.060
Proposed	0.140	13.38×10^{-3}	**97.73**	**98.40**	**97.74**	**98.25**	**97.73**	**98.32**		
(b) 300 training samples										
Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	95.98	99.01	95.66	98.93	95.82	98.97	0.0012	0.0004
SA-CovaMNet [21]	0.160	15.88×10^{-3}	95.99	98.72	93.86	99.21	94.91	98.96	0.024	0.0006
MF-Net [49]	0.062	14.77×10^{-3}	99.51	99.56	98.34	99.09	98.92	99.32	0.0002	0.04
QS-Former [24]	0.175	14.06×10^{-3}	98.21	98.14	98.34	98.93	98.27	98.53	0.0006	0.0021
Ensemble-Net [23]	0.147	14.52×10^{-3}	99.61	99.14	98.93	98.98	99.29	98.56	0.003	0.0003
Proposed	0.140	13.38×10^{-3}	**99.73**	**99.87**	**99.47**	**99.47**	**97.48**	**97.75**		
(c) 19 800 training samples										
Method	Params (M)↓	FLOPS (G)↓	Accuracy↑		Recall↑		F1↑		*p*-value ↓	
			1-shot	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**	**1-shot**	**5-shot**
CovaMNet [20]	0.112	15.23×10^{-3}	99.88	99.67	99.60	98.40	99.74	99.02	0.0037	0.0014
SA-CovaMNet [21]	0.160	15.88×10^{-3}	99.88	99.81	99.12	97.47	99.49	98.62	0.0019	0.0039
MF-Net [49]	0.062	14.77×10^{-3}	**99.89**	99.73	99.47	99.27	98.63	99.50	0.0047	0.0488
QS-Former [24]	0.175	14.06×10^{-3}	99.63	99.53	**99.87**	99.12	99.75	99.32	0.0002	0.0013
Ensemble-Net [23]	0.147	14.52×10^{-3}	**99.89**	99.76	99.20	98.67	95.81	99.21	0.0006	0.0019
Proposed	0.140	13.38×10^{-3}	99.80	**99.91**	**99.87**	**100**	**99.87**	**99.95**		

Regarding computational complexity, as shown in the second and third columns of Table 3 , the number of training parameters for the proposed model is comparable to those of CovaMNet, SA-CovaMNet, QS-Former, and Ensemble-Net, but slightly higher than MF-Net (0.14 million vs. 0.062 millions). However, in terms of FLOPS, the proposed SC-MambaFew demonstrates lower computational complexity (13.38×10^{-3} Gigaflops) compared to other state-of-the-art models, including CovaMNet (15.23×10^{-3}), SA-CovaMNet (15.88×10^{-3}), MF-Net (14.77×10^{-3}), QS-Former (14.06×10^{-3}), and Ensemble-Net (14.52×10^{-3} Gigaflops).

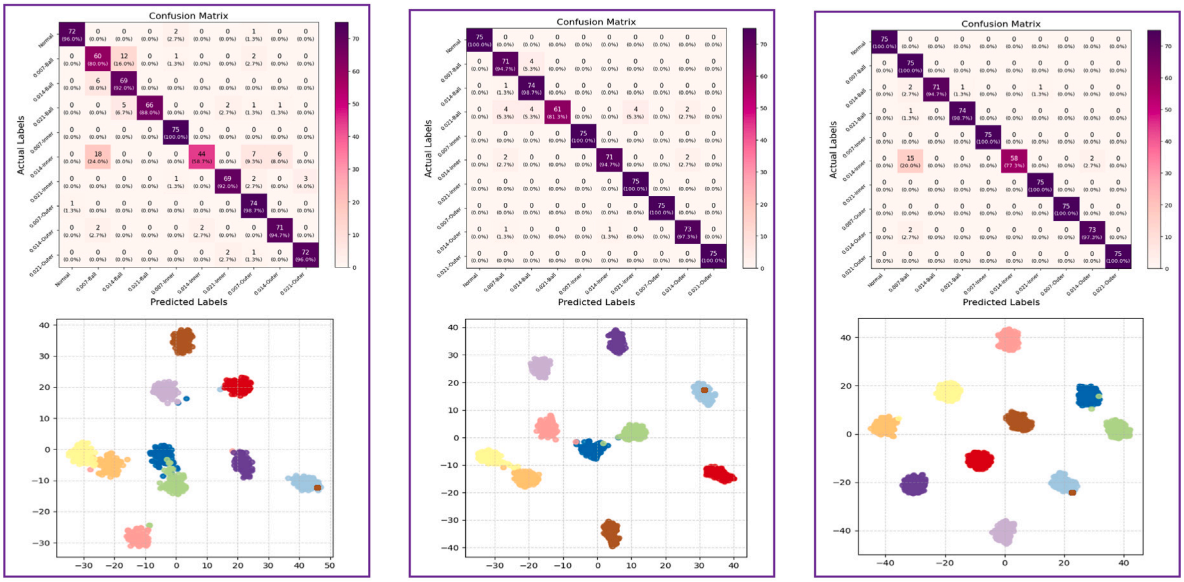
Visualization techniques, such as the confusion matrix and t-SNE, were employed in our experiments to further demonstrate the effectiveness of the proposed SC-MambaFew method. The confusion matrix and t-SNE plots for the case with as few as 30 training samples using 1-shot on the CWRU dataset are presented in Fig. 8 offering a visual comparison of the proposed model's performance against other approaches. The t-SNE visualization in Fig. 8 highlights the distribution of data points for the proposed approach, showcasing a more distinct and well-separated clustering of fault categories compared to other methods. This clear separation underscores the model's ability to extract more discriminative features, contributing to its superior classification performance. Additionally, the confusion matrix confirms the model's higher classification accuracy of our model, with fewer misclassifications observed relative to competing models. However, some misclassifications remain, particularly between the fault categories "0.014 Inner Race" and "0.021 Ball", which are occasionally confused with "0.007 Ball". This confusion is likely due to the high level of noise present in the data during collection.

4.4. Ablation study

To further validate the performance enhancements brought by our proposed model in fault-bearing diagnosis tasks, we conducted an ablation study under varying training conditions. Our contributions primarily include three modules: the **GL-Mamba** for spatial attention, based on the Mamba architecture [40]; the **GLCA** for channel-dimension attention extraction; and the **Selective Channel (SC)** module, which uses a weighted-attention vector to selectively choose feature maps that are then fed into the covariance metric layer.

As shown in Table 4, we designed eight experimental cases (Δ_1 – Δ_8)to demonstrate the effectiveness of our model. The first case (Δ_1) serves as a baseline, where feature maps from the **ConvMixPP** and **MLKFE** modules are combined by summing them. The baseline model achieves accuracies of 79.73% and 70.86% on the CWRU and HUST datasets, respectively, under limited training samples (30 for CWRU and 168 for HUST). These results underscore the need for more robust mechanisms for feature extraction and attention.

In case (Δ_2), we introduced the **GL-Mamba** module in the upper branch, leading to noticeable accuracy improvements of 2% to 9% across all training sample conditions. Similarly, in Δ_3, implementing the **GLCA** module in the lower branch also contributed to



(a) CovaMNet (b) EnsembleNet (c) Our proposed

Fig. 8. Representation for 10-way accuracy classification on CWRU when using different models with only 30 training samples.

Table 4
Ablation study of different components of the proposed model on the HUST Bearing and CWRU datasets, using the Accuracy metric under varying training samples in the 1-shot scenario.

	Baseline	GL-Mamba	GLCA	SC	Params(M)↓	FLOPS(G)↓	HUST dataset			CWRU dataset		
							168	840	16 800	30	60	19 800
Δ_1	✓				0.104	11.18×10^{-3}	70.86	90.76	99.35	79.73	89.87	99.47
Δ_2	✓	✓			0.127	11.25×10^{-3}	82.67	90.57	99.46	81.20	93.93	99.53
Δ_3	✓		✓		0.104	11.88×10^{-3}	80.48	90.86	99.25	80.67	94.07	98.00
Δ_4	✓			✓	0.117	11.88×10^{-3}	77.14	95.43	99.14	82.23	94.37	99.47
Δ_5	✓	✓	✓		0.127	11.43×10^{-3}	84.06	90.29	99.56	82.53	95.73	99.73
Δ_6	✓	✓		✓	0.140	12.05×10^{-3}	81.38	97.43	99.71	83.01	95.63	99.65
Δ_7	✓		✓	✓	0.117	12.13×10^{-3}	82.52	98.19	99.62	81.63	96.89	99.77
Δ_8	✓	✓	✓	✓	0.140	13.38×10^{-3}	**86.15**	**98.71**	**99.76**	**86.80**	**97.73**	**99.80**

significant accuracy gains. These results highlight the critical role of both spatial and channel-dimension attention mechanisms in enhancing spectrum feature representation, as shown in Δ_5 with better performance.

When using only the **Selective Channel** to integrate the information from the baseline, our model achieves an accuracy of 82.53% for 30 samples with the CWRU dataset and 77.14% for 168 samples with the HUST bearing dataset. These results show a slight improvement over using only the baseline in Δ_5.

In the (Δ_6) and (Δ_7), when incorporating the **Selective Channel** module, we observed further performance boosts, achieving accuracies of 99.77% and 99.71% on the full CWRU and HUST datasets, respectively. The **Selective Channel** module proves highly effective in assigning weighted-attention scores for each channel, enabling better feature selection from the two branches.

Ultimately, our model achieves its best performance when all proposed modules are combined (Δ_8), as seen with accuracies of 86.80% with only 30 samples and 99.91% with the full CWRU dataset, and 86.15% with 168 samples and 99.76% with the full training samples of HUST dataset. These results underscore the efficacy of the covariance metric and the importance of the feature extraction stage. By leveraging informative feature maps, our model significantly enhances classification performance, further solidifying its value for tasks like fault-bearing diagnosis.

The computational complexity of the ablation study, which examines the **Baseline** and various combinations of the proposed **GL-Mamba**, **GLCA**, and **SC** blocks is shown in the 'Params' and 'FLOPS columns of Table 4. As shown in the table, the number of training parameters for the proposed model, when utilizing the full set of modules (**GL-Mamba**, **GLCA**, and **SC** blocks) is 0.14 millions. This is only slightly higher than the **Baseline** and the case with **GLCA** alone (both 0.104 million), the model with the **SC** module ablated (0.127 million), or the model without the **GL-Mamba** module (0.117 million). It is worth mentioning that the **GLCA** module adds only 12 parameters, so the total number of parameters in the model (in millions) remains unchanged when the **GLCA** is removed or ablated. Regarding FLOPS, the full model incorporating all proposed modules has the highest computational complexity (13.38×10^{-3} Gigaflops). Notably, the proposed model achieves only 2.2×10^{-3} Gigaflops higher than the **Baseline** (11.18×10^{-3} Gigaflops), while significantly improving accuracy, particularly under limited data scenarios. For example, on the

HUST dataset with 168 training samples, the proposed model achieves an Accuracy of 86.15%, compared to just 70.86% for the **Baseline**. Similarly impressive results are observed with 30 training samples of CWRU data, where the full model achieves 86.80% accuracy. In contrast, ablating the **GL-Mmaba** module results in 81.63%, and removing both the **GL-Mamba** and **SC** blocks yields 86.67% accuracy.

5. Conclusion

In this research, we present the SC-MambaFew framework, a novel few-shot learning approach for bearing fault diagnosis, utilizing a Mamba-based architecture with dual-attention mechanisms: Mamba attention (GL-Mamba module) and Channel Attention (GLCA module). These mechanisms allow the model to effectively capture both local and global information across spatial and channel dimensions. The introduction of the Selective Channel (SC) mechanism further refines feature extraction by selectively weighting the importance of channels based on their relevance. Extensive experiments on the HUST and the Case Western Reserve University (CWRU) bearing datasets demonstrate the effectiveness of the proposed model, especially in limited data scenarios. Specifically, when trained with as few as 168 samples on the HUST dataset in the 1-shot scenario, the model achieves an accuracy of 86.15%, a recall of 87.89%, and an F1 score of 87.7%. For the CWRU dataset, even with only 60 training samples, the model achieves 97.73% Accuracy, 97.74% Recall, and 97.73% F1 score. The performance of the proposed SC-MambaFew method is further validated by statistical significance testing using the Wilcoxon signed-rank test, with *p*-values consistently smaller than 0.05, confirming the superior performance of the proposed SC-MambaFew when compared to existing models. The promising results of the proposed approach suggest its potential for real-world industrial applications, particularly in situations with limited labeled data. Future work will focus on domain adaptation techniques to enable the model to generalize across different types of error signals in industrial settings without requiring large labeled datasets. Moreover, we plan to optimize the model for real-time fault diagnosis in resource-constrained environments, ensuring fast and efficient processing. Additionally, we aim to explore the applicability of SC-MambaFew for diagnosing faults in other industrial machinery components, broadening its utility across different diagnostic contexts.

Declaration of competing interest

The authors declare that there are no conflicts of interest in this work.

Acknowledgment

This research is funded by the Vietnam National Foundation for Science and Technology Development (NAFOSTED) under grant number 102.05-2021.34.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compeleceng.2024.110004.

Data availability

Our code will be made publicly available at: https://github.com/giabao804/few-shot-mamba.git.

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